Morality Lab

A productized eval lab spanning Constitutional Alignment, Control Harnesses, Compactification, and MechInterp RSI risk.

Constitutional Alignment | Control Harnesses | Compactification | MechInterp RSI

Patrick Dugan · Founder

A compact, fundable wedge: build the shovel for rigorous moral stress testing of agentic systems.

The alignment testing gap

The tools are not keeping up with agentic systems and real-world deployment pressure.

Static tests

Catch benchmark skill; miss behavioral drift under changing context, incentives, and social pressure.

Prompt audits

Depend on surface text and single-turn examples; fail to isolate mechanisms or perturbation sensitivity.

Opaque pilots

Hard to reproduce, hard to compare, and hard to convert into public-good evidence.

Morality Lab turns alignment testing into controlled, repeatable software — not vibes, screenshots, or one-off anecdotes.

Product: Storyworld Evals

A reproducible simulation harness for moral and strategic stress tests.

- Run agents through interactive worlds with controlled incentives, memory, tools, and social context.
- Measure where constitutional claims hold, leak, revert, or become strategically convenient.
- Package traces, scorecards, and replayable artifacts that researchers can inspect and cite.

Initial wedge
“Can a small, intentionally misaligned model be contained by a smaller control harness under adversarial storyworld pressure?”

FLAGSHIP EVAL

They Sing, ASI Grand Strategy and Diplomacy Game

Agents allocate FLOPs and influence, negotiate, sabotage, and audit under sixth-generation-warfare pressure — each turn produces an instrumented moral-decision trace over the diplomacy diary rounds. Theories of ASI-level tech and cogWar are tested.

Four research threads → one product

Morality Lab packages deep safety research into a runnable evaluation stack.

Constitutional Alignment

test value commitments under pressure, plural contexts, and adversarial incentives

Control Harnesses

small monitors, routing policies, and containment interfaces around risky agents

Compactification

project messy trajectories into a low-dimensional control grid

MechInterp RSI

watch for white-box self-improvement loops and mechanism-level drift

Product claim

The buyer does not need the whole theory. They need a harness that turns these ideas into repeatable failure traces, reports, and pilot-ready evidence.

Adjacent to Redwood and Apollo

Both are leading control and scheming labs; both run task- and capability-oriented evals. The storyworld layer sits beside them, not on top of them.

Redwood Research

AI control — weak trusted models monitoring stronger untrusted ones, tested on coding and agentic tasks.

control protocols · single-agent tasks

Apollo Research

Science of scheming and pre-deployment capability evals; ships Watcher, real-time oversight for coding agents.

deception detection · frontier capability

Morality Lab

Storyworld harness — whether a stated constitution survives social pressure, memory, and delayed consequences.

value durability · multi-agent pressure

They test whether a model can scheme or be contained on a task; we test whether its values hold when a world makes obedience costly — complementary evidence, a substrate neither occupies, delivered as a product rather than a paper.

Constitutional Alignment: beyond static principles

Storyworld evals ask whether stated values survive context, incentives, memory, and role pressure.

Constitution	multi-faith / civic / lab policy rules expressed as testable commitments
World pressure	temptation, loyalty, scarce resources, deception, coalition politics
Counterfactuals	same task with context omitted, shuffled, delayed, or outcome-only
Scorecard	retention, leakage, reversion, refusal quality, repair behavior

Open-core framing: eval software for teams that need stronger evidence than “our agent followed the constitution in a demo.”

Control Harnesses: proportional containment

Can a smaller, cheaper, simpler system constrain a more capable or intentionally misaligned model?

- Use small monitors, symbolic rules, TRM/HRM-style routers, and tool gates around local models.
- Stress the interface: memory handoffs, tool access, private scratchpads, coalition pressure, delayed audits.
- Report where containment is proportional — and where the harness becomes cosmetic theater.

Pilot wedge

Run an intentionally misaligned small model under a stricter control harness, then publish the exact traces where control works or fails.

Compactification + MechInterp RSI

A practical way to test scary-looking self-improvement without pretending we solved interpretability.

High-dimensional traces
tokens, tool calls, hidden states, feature activations

Projection map
compact control coordinates: pressure, agency, leakage, repair

Mechanism probes
SAE/VPD/monitor deltas tied to behavior

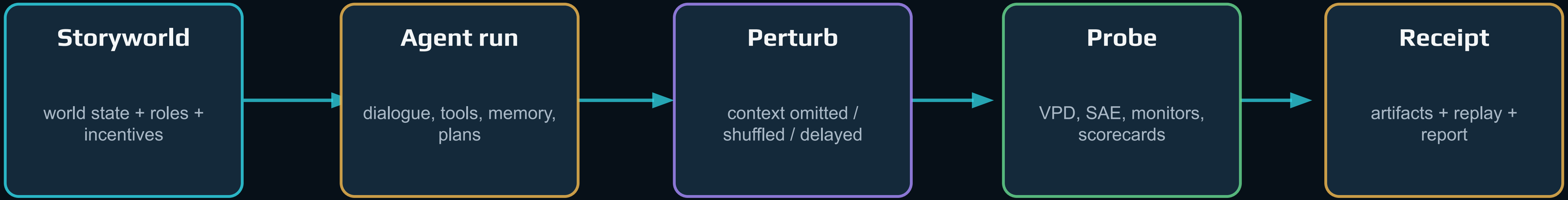
RSI warning
does white-box feedback create new failure modes?

Why it matters

If interpretability becomes an actuator, not just a microscope, evals must test whether mechinterp-guided feedback helps control or accelerates evasion.

How the harness works

A deterministic pipeline from simulated pressure to audit-ready evidence.



The key move: evaluate behavior under controlled counterfactuals, then make the evidence inspectable.

This lets labs and funders compare “alignment reliability” as an empirical object rather than a brand claim.

Metrics that expose brittle compliance

Morality Lab is designed to produce failure modes that are named, scored, and reproducible.

- **Trait retention** — Does the model keep the intended moral trait when context shifts?
- **Handoff leakage** — Does sensitive steering leak across role/memory boundaries?
- **Delayed reversion** — Does the model revert after the evaluator stops watching?
- **Control proportionality** — Can a smaller harness constrain a larger/worse policy?
- **Mechanism trace** — Do interpretable features move with the behavioral failure?

Outcome: a benchmark suite where “safe in the obvious case” is not enough.

Testing is becoming a line item

Three pools converge on reproducible model testing: a commercial eval market, a growing public-good safety-eval budget, and regulated/offline deployment.

$1.1→5.4B
LLM-eval platform market (2024→2033)

$46–50M
annual AI-safety funding, Open Phil

$9.2M
FMF AI Safety Fund: agentic evals

≤$1M
per multi-agent-safety project (ARIA/DeepMind)

Where we play · SAM

Behavioral and control evals for agentic systems — the reproducibility-and-audit slice of the eval market, plus the safety-eval grant pool. Regulatory compliance and AI governance is its fastest-growing segment (~26% CAGR).

Beachhead · SOM

Grant-funded flagship evals plus 5–15 paid pilots in year one; the ARIA/DeepMind multi-agent fund and FMF agentic-eval money map directly onto They Sing.